

DepthWizard

Single-View Height Estimation and Interactive 3D Flythrough from Satellite Imagery

Technical Documentation and AI Handoff Guide

Problem Statement ID: 26175 | Organization: ISRO | Theme: Disaster Management

1. Executive Summary

DepthWizard is an end-to-end software pipeline that accepts a single optical RGB remote-sensing image and produces: (1) an AI-derived relative depth representation, (2) a Relative DSM for non-georeferenced imagery or an estimated calibrated DSM for georeferenced imagery, and (3) an interactive 3D environment that supports inspection and flythrough.

The project combines pre-trained and open-source components for rapid prototyping. Depth Anything V2 is used primarily for relative geometric cues and structural analysis. Sat3DGen is used primarily for satellite-image-based 3D scene reconstruction and mesh generation. Rasterio handles GeoTIFF metadata. A proposed GeoScale module aligns relative AI geometry with coarse absolute elevation references such as SRTM or limited Ground Control Points. Three.js provides interactive visualization.

2. Core Problem

Traditional DEM/DSM acquisition can require stereo imagery, LiDAR, or InSAR. These approaches may depend on specialized sensors, multiple acquisitions, cost, and computationally intensive processing. The challenge is to obtain useful geometric and elevation information from a single RGB image while clearly distinguishing relative depth from true metric elevation.

Critical scientific constraint: a generic monocular depth model does not automatically produce survey-grade elevation in meters from a satellite image. Metric elevation requires calibration and validation against reference information.

3. Required Inputs and Outputs

Non-Georeferenced Input: JPG, JPEG, PNG, or TIFF without usable spatial metadata. Outputs: relative depth map, Relative Digital Surface Model (rDSM), visualizations, 3D mesh/scene where reconstruction succeeds, and relative height/slope analysis.

Georeferenced Input: GeoTIFF or imagery containing usable spatial metadata. Outputs: relative depth map, extracted CRS/bounds/transform/resolution, aligned reference DEM, estimated calibrated DSM, slope products, and interactive 3D visualization.

4. Essential Terminology

DEM: Digital Elevation Model. Usually represents bare-earth terrain elevation, ideally excluding buildings and vegetation.

DSM: Digital Surface Model. Represents the top surface, including terrain plus buildings, trees, and other objects.

rDSM: Relative DSM. Encodes relative surface variation but is not guaranteed to be expressed in meters.

GeoTIFF: A TIFF raster containing geospatial metadata such as coordinate reference system and georeferencing transform.

CRS: Coordinate Reference System. Defines how coordinates correspond to locations on Earth.

GCP: Ground Control Point. A known real-world location/value used for geometric or scale calibration.

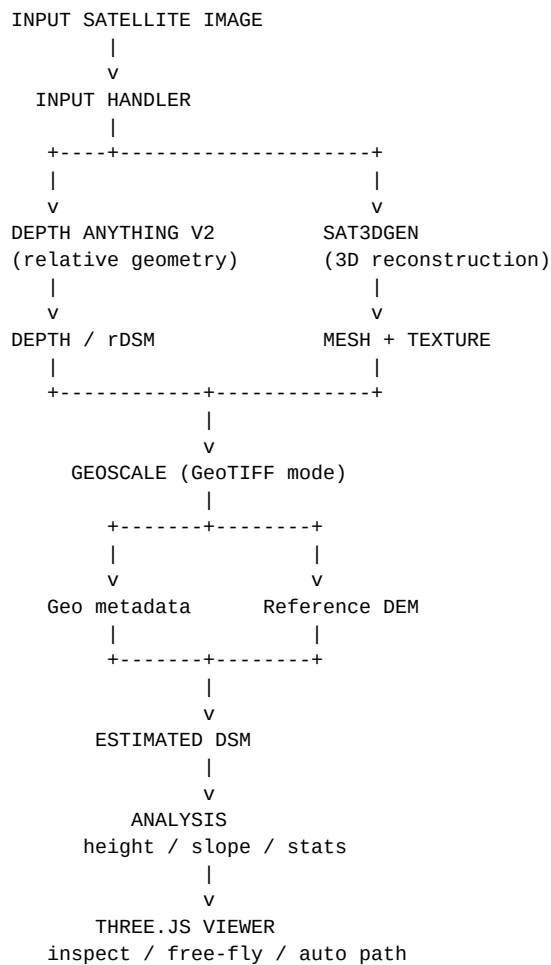
SRTM: Shuttle Radar Topography Mission data; commonly used as a coarse terrain elevation reference.

Relative depth: Depth values meaningful mainly in relation to other pixels rather than as absolute metric distances.

Mesh: A 3D surface made from vertices, edges, and usually triangular faces.

Texture: The image/material mapped onto the surface of a 3D mesh.

5. Final System Architecture



6. Component Responsibilities

6.1 Input Processing Module

Detect file type, validate the raster, convert imagery to RGB where needed, and determine whether geospatial metadata is available. For GeoTIFFs, extract CRS, affine transform, bounds, dimensions, and pixel resolution.

6.2 Depth Estimation Module: Depth Anything V2

Input: RGB image. Output: a relative depth representation. Normalize the output and verify its depth convention. Use it for relative structure and local surface variation; do not present it as direct metric elevation without calibration.

6.3 3D Reconstruction Module: Sat3DGen

Input: satellite image. Role: generate a 3D scene representation and exportable mesh assets supported by the repository's actual inference pipeline. Do not assume a specific output format; inspect the real generated artifacts.

6.4 GeoScale Module

Project integration module that maps scale-agnostic AI geometry toward absolute elevation when geospatial references are available. Inputs may include relative depth, aligned coarse DEM, and optionally GCPs.

6.5 Analysis Module

Provides point queries, relative or calibrated elevation display depending on mode, slope calculation, and summary statistics.

6.6 Visualization Module

Loads the mesh into a Three.js scene and provides camera, lighting, sky/background, inspect controls, free-fly navigation, and automatic flythrough.

7. GeoScale Calibration Logic

The simplest prototype should use an explicit, defensible calibration rather than claiming a sophisticated method that has not been implemented.

```
D = normalized relative depth
D_low = smooth(D)
D_detail = D - D_low
DEM_aligned = reference DEM resampled/reprojected to target grid
```

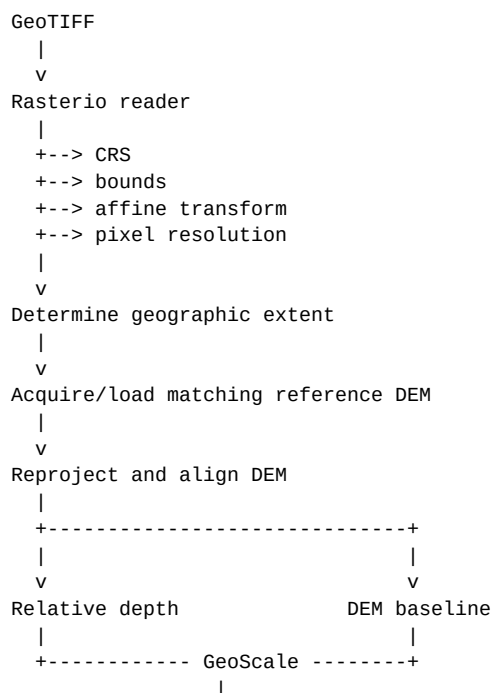
```
Prototype fusion:
DSM_est = DEM_aligned + alpha * D_detail
```

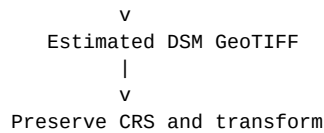
```
Alternative when reference points exist:
Z = a * D + b
```

Estimate a and b using regression on valid reference samples.

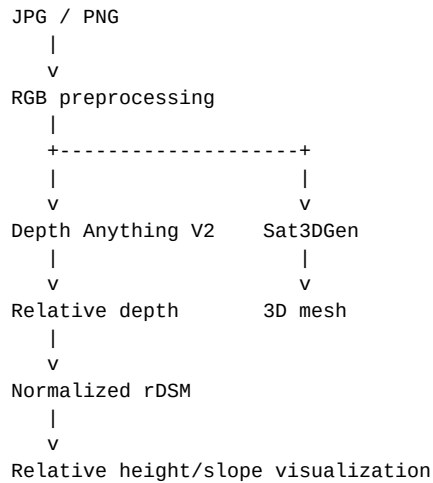
The calibration factor alpha, or regression parameters a and b, must be estimated from available reference data and validated. Coarse DEMs provide broad terrain trends; they generally cannot validate individual building heights at high precision.

8. Georeferenced Processing Pipeline



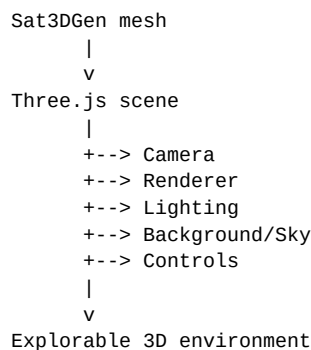


9. Non-Georeferenced Processing Pipeline



10. 3D World and Flythrough

A mesh is not a complete interactive application by itself. The mesh becomes the principal geometry of the 3D world after being loaded into a rendering scene.



Inspect Mode: orbit-style controls for rotating, zooming, and inspecting the reconstructed area.

Free-Fly / Drone Mode: W/S forward/backward, A/D lateral movement, Space upward, Shift downward, and mouse for look direction. Collision can be omitted for the MVP because the use case is aerial exploration.

Automatic Flythrough: define camera waypoints and interpolate camera position and target direction. A pre-rendered trajectory video can be used as a fallback demonstration.

11. Suggested Folder Structure

```

DepthWizard/
  README.md
  requirements.txt
  app.py
  core/
    input_handler.py
    depth_engine.py
    reconstruction.py
    geoscale.py
    pipeline.py
  geospatial/
    geotiff_reader.py

```

```

■ ■■■ dem_loader.py
■ ■■■ alignment.py
■ ■■■ exporter.py
■■■ analysis/
■ ■■■ height.py
■ ■■■ slope.py
■ ■■■ metrics.py
■■■ frontend/
■ ■■■ index.html
■ ■■■ main.js
■ ■■■ viewer/
■ ■■■ mesh_loader.js
■ ■■■ fly_controls.js
■ ■■■ flythrough.js
■■■ data/
■ ■■■ input/
■ ■■■ dem/
■■■ outputs/
■■■ depth/
■■■ rdsm/
■■■ dsm/
■■■ mesh/
■■■ video/

```

12. Technology Stack

Python: orchestration and geospatial processing.

PyTorch: AI inference.

Depth Anything V2: relative monocular geometry.

Sat3DGen: satellite-to-3D reconstruction.

Rasterio: GeoTIFF reading/writing.

NumPy/OpenCV/SciPy: array and image processing.

Three.js: browser-based interactive 3D rendering.

Gradio or FastAPI: rapid application interface/backend.

13. Evaluation

The problem statement emphasizes both elevation accuracy and visualization quality.

```

RMSE = sqrt( mean( (Prediction - GroundTruth)^2 ) )
MAE = mean( abs(Prediction - GroundTruth) )
Correlation = statistical relationship between predicted and reference values

```

Before computing metrics, predicted and reference rasters must be aligned spatially, use compatible vertical units/datums where applicable, and exclude invalid/no-data pixels. Evaluate separately where possible across urban, sparse, hilly, and forested landscapes.

Visualization: assess mesh loading, texture quality, navigability, camera smoothness, UI clarity, rendering stability, and successful deployment.

14. Honest Claims for Presentation

Safe claim: DepthWizard estimates relative geometry from a single image and uses available geospatial references to calibrate elevation where possible.

Unsafe claim: Depth Anything V2 directly generates LiDAR-accurate building heights from any satellite image.

Safe claim: The project integrates complementary pretrained reconstruction and depth-estimation systems into a unified remote-sensing workflow.

15. One-Day MVP Plan

PRIORITY 1: Make one end-to-end demo work.

Image -> Depth Anything V2 -> depth visualization

PRIORITY 2: Obtain one real Sat3DGen reconstruction.

Image -> Sat3DGen -> mesh

PRIORITY 3: Build the viewer.

Mesh -> Three.js -> inspect + free-fly

PRIORITY 4: Add geospatial demonstration.

GeoTIFF -> metadata -> preloaded DEM -> simple calibration

PRIORITY 5: Integrate UI and outputs.

Do not block the entire demo on perfect metric calibration.

Prepare precomputed outputs for at least one reliable demo scene. If a live model fails or is too slow, the application should still demonstrate the complete pipeline with previously generated results.

16. Expected Outputs

Non-georeferenced: original image, depth visualization, relative depth/rDSM data, slope visualization, mesh, and optional flythrough video.

Georeferenced: extracted metadata, reference DEM, estimated DSM GeoTIFF preserving spatial metadata, slope GeoTIFF, mesh/3D visualization asset, and validation metrics where reference ground truth exists.

17. Final Elevator Pitch

DepthWizard is an AI-assisted remote-sensing framework that transforms a single optical satellite image into relative geometric structure and an explorable 3D environment. For georeferenced imagery, it combines AI-derived structural cues with available spatial metadata and coarse elevation references to estimate a calibrated DSM. The resulting reconstruction can be explored through interactive inspection, free-fly navigation, and automated flythrough while supporting height and slope analysis.

18. Instructions for Another AI Working on This Project

Treat this document as architectural intent, not proof that every repository component is directly compatible. First inspect the current repositories, model requirements, licenses, input/output formats, hardware needs, and actual generated artifacts. Preserve the separation of responsibilities: Depth Anything V2 for relative geometry; Sat3DGen for 3D reconstruction; GeoScale as the integration/calibration layer; Rasterio for geospatial metadata and raster export; Three.js for interactive visualization. Prefer a functioning end-to-end MVP over implementing every advanced idea.

Most important implementation rule: never silently convert relative depth values into meters. Only display metric elevation after an explicit calibration procedure and label estimated outputs clearly.